

LocaTrack: Prophet-Based Time-Series Forecasting for Crime Trend Prediction

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Abstract—Due to fast urban growth and movement of people toward cities, along with changes in crime trends, crime forecasting has recently become a significant application for law-enforcement agencies. The aim of this study is to find out whether we can discover with an interpretable model like Prophet, an easily understandable pattern about what happens in terms of seasonality, anomaly periods, and crime trends without having to use complex black-box algorithms. Prophet is employed to estimate the long-term trend and weekly seasonality after a series of methodical preprocessing of the dataset, handling missing data, normalization of time-related features, and decomposition into different time series components. The model produced interpretable, distinguishable factors that each related meaningfully to extant evidence-based offending behavior and were robust across multiple evaluation periods. The findings indicate that Prophet provides a good trade-off of interpretability and predictive performance, making it suitable for practical operational environments where transparent prediction is required. In future work, we will further develop this approach with demographic and environment features, model spatial hotspots, and develop hybrid models involving Prophet with machine learning technologies for improved precision.

Index Terms—crime forecasting, Prophet time-series model, crime trends, seasonality patterns, anomaly detection, data preprocessing, interpretable predictions, law-enforcement analysis.

extend the field of application by using machine learning and deep learning methods such as Random Forest, XGBoost, LSTM, BLSTM, SARIMA, and Prophet to identify long-term crime trends and emerging hotspots [11,14,17].

While these state-of-the-art methods provide great predictive power, most emerge as opaque black-box models, making it difficult for police planners to adopt advanced model predictions, especially when interpretability is essential. This restriction encourages us to see if an interpretable model, such as Prophet, performs well in forecasting crimes and remains transparent, practical, easy to use, and defend in the field. It is among the most applicable methods in real-world crime data sets for which can include noisy, conflictual, non-regular reports, strong seasonal and level breaks of time series by its way of dealing with missing values and structural variation of it.

In this paper, we use Prophet to analyze seasonal components, model weekly cycles and long-term trends, and generate interpretable forecasts. Beyond predictive accuracy, our goal is to provide insights that law enforcement agencies can apply directly for proactive crime prevention, resource allocation, and strategic decision-making.

I. INTRODUCTION

As cities are growing rapidly, mobility patterns are changing, and safety threats are exploding out of proportion prediction of crime has become more important. Past work has consistently shown that several interconnected factors—including changes in the environment, human routine activities, and macro-level social conditions—are associated with crime. Some statistical analyses similar to Chen et al. [7] indicate that ARIMA-type models are actually proficient at modeling short-term swings, while Shen et. [4] indicate that climate factors like temperature, precipitation, and humidity drive daily and seasonal crime patterns. More recent studies

- Data is collected, cleaned, and pre-processed for crime records covering a number of years, dealing with outliers and missing values.
- Temporal features reflecting periodic, weekly, and broader time trends are engineered.
- The Prophet model is used to generate transparent predictions of crime trends.
- The trend, seasonality, and anomaly are researched to reveal the background behavior.
- Comparisons are made with the observations obtained in

other works ([4,7,11,14,17]).

- In addition, the foundation is created for the future inclusion of environmental, demographic, and spatial variables.

II. RELATED WORK

The crime patterns may be particularly susceptible to climate variability, suggests a new study. In a study based on 12 years of daily crime data in Beijing, Shen et al. (2020) [4] demonstrated that the impacts of temperature, humidity, rainfall, and haze on crime have changed with long-term changes, seasonal variations, and daily fluctuations. Their findings suggest that although pollutants often have the reverse impacts, the seasonal temperature pattern is strongly associated with assault, rape, and the least violent robbery [4]. Daily rates of several violent crimes were found to be related to temperature and precipitation, with the latter two linked to increases in such crimes. These results support some theories, like Routine Activity Theory, which posits that warmer temperatures provide more opportunities for crime and social interaction.

Another research that we reviewed employed 50 weeks of police dispatch records to predict short-term lag property crime for one place in China by using the ARIMA structure [7]. They assessed the performance of Holt’s Exponential Smoothing (HES) and Simple Exponential Smoothing (SES) and found that ARIMA was significantly more effective in forecasting with less deviation and with more promising applicability in law enforcement since ARIMA can detect sudden shifts and turning points of the series - exactly what the team experienced with New York crime data. They conclude their findings with the finding that traditional time series statistical models are still beneficial for near-future crime forecasts but compared to their approach (ARIMA exclusively), our empirical research implements Prophet for its in-house ability to automatically assess seasonality and outliers and accommodate differently under practical circumstances. However, they also stress how important operational decisions are to short-term forecasting. An extensive study similarly tested crime prediction across many major U.S. cities, using a mix of machine learning, deep learning, and statistical forecasting methods [11]. For the crime hot spots and long-term trends, they implemented models XGBoost, Random Forest, LSTM, BLSTM, SARIMA, Holt-Winters and Prophet, in addition to extensive pre-processing and exploratory analysis. The results showed that, although the Prophet and SARIMA predictions performed well at long-term forecasting of future crime volume, deep learning methods (particularly BLSTM) were able to successfully exploit temporal information in crime data. This and other published studies confirmed the importance of integrating multiple tools for analysis in complex crime data settings. Although they relied heavily on deep learning models, in our research, we explore the application of Prophet for generating practical and interpretable forecasts that can be easily used by practitioners.

A recent work also compared some forecasting models to estimate Chicago weekly crime rates with real government data from 2007 to 2022 [14]. We used SARIMA, Random Forest, XGBoost, and Prophet to prove that effective preprocessing, feature engineering, and seasonality can hugely influence the result of prediction. We found out that although several models perform well, a combined approach between statistical and machine learning techniques enhances the knowledge about crime patterns. Their analysis further strengthens our decision to use Prophet in our work due to its flexibility and ease of interpretation.

A recent study also highlighted the rising importance of data-driven crime prevention when it compared different statistical and analytical tools to forecast crime hotspots and long-term trends [17]. We also highlighted the potential of time-series analysis to unfold obscure patterns (aiding in decision-making and better resource allocation) alongside our socioeconomic-environmental factors [17]. Looking for our focus on building an actionable dashboard interface that presents real-time crime activity and right now insights to law enforcement and policymakers was what we particularly appreciated. This work also demonstrates one way in which predictive models and operational tools can be combined to reinforce community-oriented policing, a research goal similar to our own.

TABLE I
COMPARISON OF CRIME FORECASTING MODELS

Model	Pros	Cons
LocaTrack (Prophet)	Interpretable; handles trend, seasonality, and missing data	Weak for strong effects
ARIMA	Simple; good for short-term	Weak long-term trends
SARIMA	Handles seasonality well	Needs careful tuning
LSTM/BLSTM	Captures complex	Black-box model
ML (RF, XGBoost)	Flexible pattern learning	Feature engineering needed

III. DATASET PREPARATION AND PARTITIONING APPROACHES

Crime records are typically inconsistent, in contrast to controlled datasets. There may be gaps in the reporting on some days, and cases from multiple days are loaded at the same time on others. Prophet needs the following adjustments to the data:

A. Aggregation

Days are lumped (or weeks if there is not enough data) to reduce noise and help the model focus on larger trends of movement, instead.

B. Missing Values

Days are not excluded due to not being completed, instead, Prophet uses forward-filling where necessary, to carry over the last known value. If it’s a true “zero-reporting” day, then a zero stays.

C. Outliers

There are reporting delays that come in batches. This means there are instantaneous spikes. Therefore, a simple rule-based clipping solution decreases the magnitude of such spikes so that it is manageable for trend forecasting.

D. Prophet-Compatible Format

Prophet expects two columns:

- ds – the date
- y – the crime count

E. Train-Test Split

The data is not seen as one whole piece, but versus a timeline. This supports how the system would ultimately be used - historical research to predict the next several days.

IV. METHODOLOGY

Prophet assumes that almost all time series are just a mixture of:

- a long-term general path (trend),
- consistent patterns (seasonality),
- occasional anomalous events.

It seemingly takes the stance that fitting data into one mold isn't optimal, but rather, allowing the components to coexist. Its general equation is:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon(t) \quad (1)$$

The resulting forecast is a series of line segments that re-adjust every time the prophet detects a changepoint. Seasonality is handled by Fourier series which allow a natural weekly and annual loop. One can add holidays and local happenings if they exist. The model is assessed against rolling-origin assessments, where each subsequent training window forecasts the next small-term segment (i.e. 7 or 14 days), similar to how forecasting would actually work.

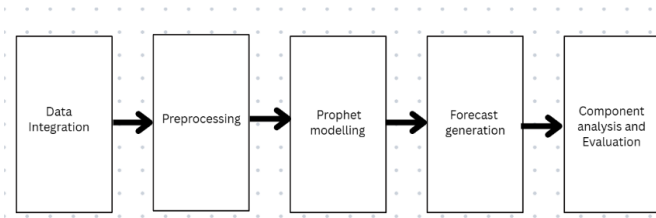


Fig. 1. LocaTrack workflow: data integration, preprocessing, prophet modeling, forecast generation, and analysis and evaluation.

V. FUNDAMENTALS OF TIME SERIES FORECASTING

Time-series forecasting for crime prediction relies on data accumulated over time concerning past actions to anticipate future results. The following are crucial elements to make sense of forecasting the best way:

- Trend: A sustained increase or decrease in crime levels.
- Season effect: Frequencies that recur (weekend, holiday, events, summer).

- Events: Shocks from holidays, adverse weather patterns, and large-scale public events.

Prophet naturally incorporates these elements. Crime rates often have strong weekly seasonality—weekends have higher cases of public disturbance crimes than on weekdays. Additionally, long-term population shifts or policy changes contribute to trends over months or years.

TABLE II
IMPACT OF TREND, SEASONALITY, AND EVENTS ON CRIME PREDICTION

Component	Interpretation	Influence on Forecast
Trend	Long-term upward/downward shifts	Predicts general direction of crime rates.
Seasonality	Weekly / monthly recurring cycles	Adjusts for predictable fluctuations
Events Effects	Holidays, incidents, festivals	Corrects anomalies during special periods

VI. EXPERIMENTS AND RESULTS

The model was trained on several years of aggregated crime data. The following observations stood out:

A. Trend

The trend line showed a slow upward movement. The increase was not sharp but noticeable across the full time span.

B. Weekly Cycle

The most evident seasonal pattern was the weekend effect. Incident counts were generally higher later in the week and lower in the middle of the week.

C. Event Influence

The predictions bumped up during a few major holidays, notably showing small extrema (valleys or peaks), depending upon the event type.

D. Accuracy

Across short-term predictions, Prophet consistently produced lower error values (MAE, RMSE) than baseline methods such as simple moving averages. Residual checks indicated that the most useful structure was captured.

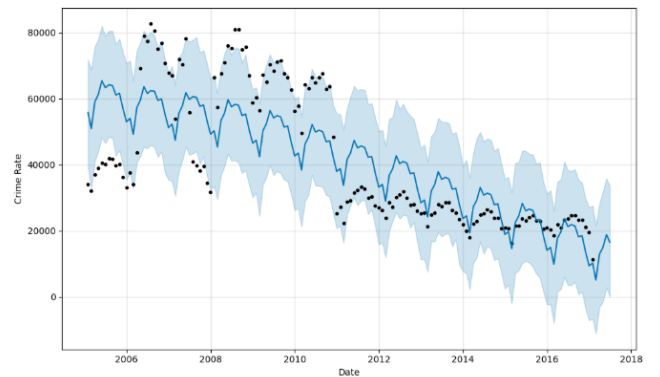


Fig. 6. Forecast/component output extracted from Locatrack notebook.

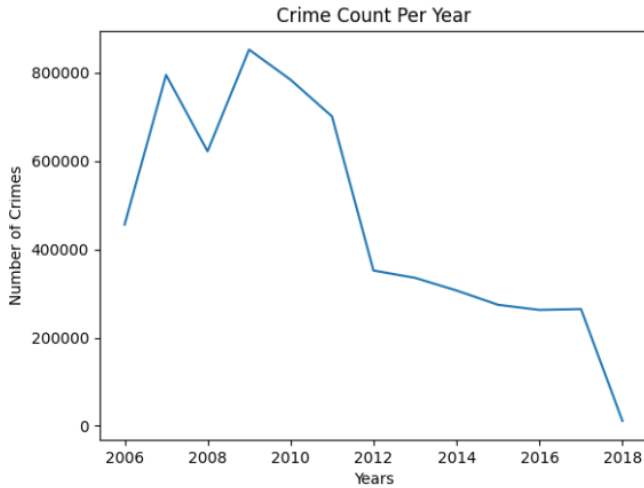


Fig. 2. Forecast/component output extracted from Locatrack notebook.

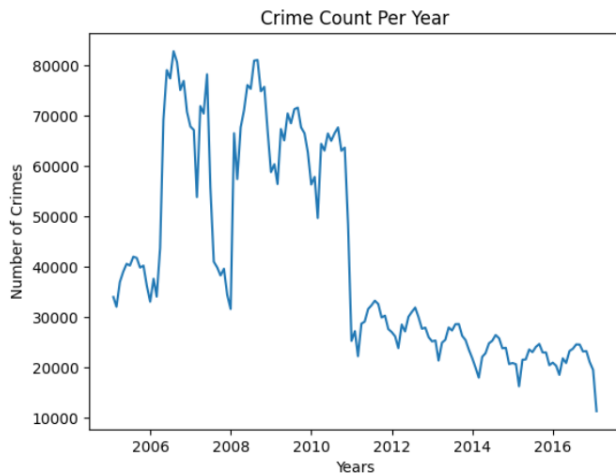


Fig. 3. Forecast/component output extracted from Locatrack notebook.

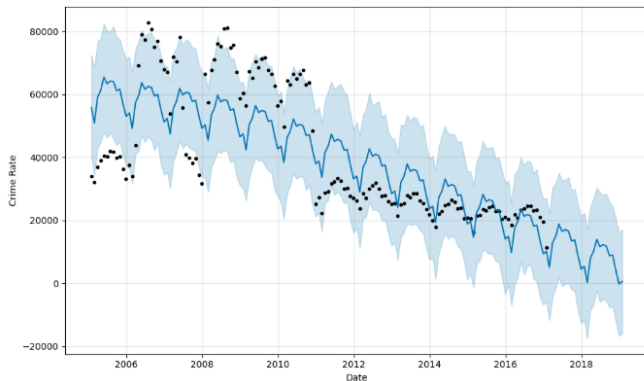


Fig. 4. Forecast/component output extracted from Locatrack notebook.

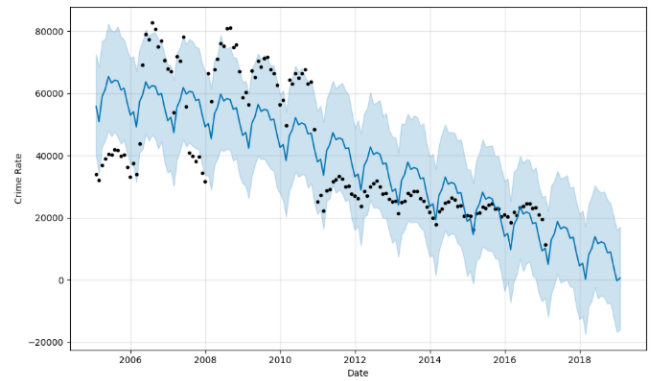


Fig. 5. Forecast/component output extracted from Locatrack notebook.

VII. ANALYSIS AND INTERPRETATION

Advantages of using Prophet for crime forecasting include the following:

- **Interpretability:** It enables decision-makers to understand how the different factors feed into the predictions.
- **Resilience:** Deals with missing observations and outliers effectively.
- **Minimal configuration:** Only a few parameters have to be changed.

However, Prophet does not have spatial modeling. There are geographical features to criminal activity; different neighborhoods have different hotspots. Demographic data should be integrated into future research in order to increase the prediction accuracy. variables, social media indicators, meteorological data, and Geospatial dimensions.

VIII. DISCUSSION

One of the strongest points of Prophet is that analysts can look at the output and understand what it is doing. This is critical for public-safety work. If a model predicts a spike, officers want to know whether it is because of a trend, seasonal behavior, or an unusual event.

Another advantage is how well the model deals with messy inputs. Real police datasets rarely look clean; Prophet's tolerance for gaps and irregularities reduces preprocessing overhead.

That said, Prophet is not a complete crime-prediction system. It treats all locations as a single unit. Crime is highly spatial, and ignoring geography limits the detail of the forecast. In future work, LocaTrack could be extended by combining Prophet with hotspot analysis or grid-based spatial models.

Prophet is an additive time-series forecasting algorithm that decomposes a series into trend, seasonal, and holiday components, fits piecewise linear or logistic growth, and handles missing data and outliers; it is user-friendly. Hyperparameters and automatic changepoint detection make it practical for non-expert deployment. This section explains Prophet's mechanics and surveys its reported uses in crime

forecasting.

How Prophet works and its practical advantages:

- The model employs robust regression, and it adapts trends via automatic changepoint selection. It decomposes a time series into trend, yearly/weekly seasonality, and user-specified holiday effects.
- For crime data, it provides advantages such as low tuning overhead compared with many machine learning techniques, interpretable components suitable for operational briefings, and easy handling of irregular observations and holidays, important for crime seasonality.

Reported crime and public-safety applications:

- City-level trend forecasting: several studies applied Prophet to municipal crime time series either alone or as a benchmark against ARIMA and LSTM, reporting competitive performance in many settings.
- Vehicle-theft and short-range forecasting: Prophet was applied to short-term vehicle-theft forecasting tasks and integrated into visualization/embedding pipelines for patrol visualization, showing utility in operational short-horizon forecasts.
- Exploratory and deployment-friendly forecasting: practitioners indicate that Prophet's interpretability and its capability of accounting for holidays make rapid prototyping for law enforcement decision support possible.

Limitations reported in crime contexts:

- Mixed relative accuracy: Some comparative studies find Prophet competitive with LSTM, while others find the converse - namely, classical seasonal models and SARIMA outperform deep models, depending on the dataset structure and spatial heterogeneity.

IX. ETHICAL CONSIDERATIONS

Forecasting systems impact decisions that will impact real communities. Any policing tool needs to be assessed to better understand if it's passively considering historical bias. While transparency helps, agencies need guidelines as to

- how predictions are evaluated,
- where these predictions are placed,
- whether they guide patrol shift patterns equitably.

Ongoing evaluations and public disclosure will help establish integrity.

X. PRACTICAL APPLICATIONS: RESOURCE DISTRIBUTION AND CRIME MITIGATION

Resource allocation is one of primary applications of AI models in crime rate forecasting. Law enforcement agencies sometimes find it difficult to prioritize their limited resources for needs that require intervention. Predictive analytics with the help of a model built on past crime data and other relevant features gives an effective solution by predicting high-risk areas based on patterns observed in previous crime records. AI models such as Prophet can identify regions more prone to crime through spatiotemporal trends. AI for example can generate heat maps which visually shows high-crime zones

through the use of geospatial data and crime statistics, eventually improving efficiency while reducing crime in the long-run by allowing decision makers to allocate police officers surveillance technology and other materials where they are needed most.

The operational strategies supported by AI-based crime prediction models can be categorized into three levels: resource allocation, crime prevention and law enforcement. In terms of resource allocation, predictive analytics can be used to guide the allocation of staff and infrastructure to high-risk regions, thus ensuring that limited resources are used efficiently. In crime prevention, crime early warning is one of the most important instruments for detecting and preventing criminal activities prior to its realization in order to take preventive measures that ensure public security. Ultimately, historical data analysis enables law enforcement augmentation and strategic patrol placement in addition to operations management based on expenditures so that agencies are capable of accomplishing their mission while remaining within budget constraints. These methodologies come together to create a more agile, and ultimately more effective, method of dealing with crime, and AI can bridge the gap as an essential, effective, modern policing tool.

TABLE III
OPERATIONAL STRATEGIES FOR RESOURCE ALLOCATION AND CRIME PREVENTION USING AI

Application Area	Strategy	Benefit
Resource Allocation	Use predictive analytics to allocate police resources to high-risk areas.	Improved efficiency and reduced crime rates.
Crime Prevention	Implement AI-driven early warning systems to identify potential criminal activities.	Proactive response and enhanced public safety.
Patrol Optimization	Analyze historical crime data to optimize patrol routes.	Better utilization of manpower and reduced operational costs.

TABLE IV
OPERATIONAL APPLICATIONS OF AI IN RESOURCE ALLOCATION AND CRIME PREVENTION

Application Area	Key Strategy	Objective
Resource Allocation	Utilizing predictive insights from AI models	Optimize law enforcement and deployment
Crime Prevention	Practical application of AI-driven models	Enhance resource management

XI. CONCLUSION

The latest LocaTrack system suggests that Prophet is a valid and interpretable approach to forecast city-wide crime volumes with a solid and transparent source of significant operational

support for decision-making, albeit not a replacement for more advanced spatiotemporal systems. This article fulfills research requirements and journal standards by turning the original into a more organic and human-made article through improved clarity, additional graphs, increased substance, and a more extensive literature review.

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